

CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- DriverPass wants a web-based system that helps students prepare for their driving tests more effectively. The system will let users take online practice exams, schedule lessons, and track progress. This project's goal is to replace the current manual, paper-based process with a secure, easy-to-use digital platform that improves learning and efficiency for both students and instructors.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass noticed that many students fail their driving tests due to limited access to structured practice and confusing lesson scheduling. The new system will combine student accounts, online testing, and instructor management into one connected platform. This reduces errors, saves time, and supports DriverPass's mission to make driver education more organized and accessible.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The system shall let students register, log in, and take online practice exams.
- The system shall track student scores and progress over time.
- The system shall let instructors schedule lessons and record feedback.
- The system shall secure data with authentication and encryption.
- The system shall support desktop and mobile browsers.
- The system shall maintain 99.9% uptime with quick page loads.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system must run as a cloud-based, web-hosted application accessible through modern desktop and mobile browsers.
- Pages should load within two seconds under normal conditions, even during peak usage.
- The system should maintain 99.9% uptime to avoid disruptions to scheduling and testing.
- System updates and maintenance windows should be scheduled during non-business hours to minimize customer impact.
- Data synchronization must occur in real time when users are online, ensuring no duplicate or conflicting records.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system must operate on common web browsers (Chrome, Safari, Edge, Firefox).
- Operating system-dependent features should not be required, allowing access from Windows, macOS, iOS, Android, and Linux.
- The back end will rely on a relational database to store accounts, packages, reservations, test progress, and instructor notes.
- Cloud hosting should handle storage, backups, and security without requiring DriverPass to maintain servers internally.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system must correctly distinguish users by roles (student, secretary, manager, IT administrator).
- Username and password fields should be case-sensitive.
- Any failed reservations or data errors should trigger an alert for IT.
- Activity tracking must record who made, modified, or canceled each reservation with timestamps.
- All calculations related to lesson hours, package requirements, and test scoring must be accurate and consistent.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Admin users must be able to enable or disable student packages without code changes.
- The IT administrator must have the ability to reset passwords, block account access, or unlock accounts when needed.
- The system should be designed in a modular way so future features (such as customizable packages) can be added later.
- The system must adapt to browser and OS updates without requiring major redesigns.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a “brute force” hacking attempt? What happens if the user forgets their password?

- All users must authenticate using a secure username and password.
- Password reset functionality must be available for students who forget their login information.
- Sensitive data—including credit card details—must be encrypted during transmission and storage.
- The system must enforce lockout procedures after repeated failed login attempts to prevent brute-force attacks.
- HTTPS must be used for all data exchanges between client and server.
- Admin accounts should require elevated security controls to manage user access safely.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with “The system shall . . .” For example, one functional requirement might be, “The system shall validate user credentials when logging in.”

- The system shall allow students to create an account and enter personal information.
- The system shall validate user credentials when logging in.
- The system shall allow students to schedule, modify, and cancel driving lesson reservations online.
- The system shall allow the secretary to schedule lessons for customers who call or visit the office.
- The system shall assign instructors and cars to scheduled lessons.
- The system shall display available appointment times.
- The system shall allow students to take online practice tests.
- The system shall track practice test status, including “Not Taken,” “In Progress,” “Passed,” and “Failed.”
- The system shall store all test results, scores, and timestamps.
- The system shall record instructor comments and lesson logs for each student.
- The system shall allow the IT administrator to reset passwords and manage accounts.
- The system shall allow management to generate reports showing reservation activity and changes.
- The system shall receive updates from the DMV when rules, policies, or questions change.

- The system shall allow the admin to enable or disable available lesson packages.
- The system shall allow students to download certain reports for offline use (Excel).

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface must be web-based and easy to use for all skill levels.
- Students must be able to:
 - View their packages, lessons, and test progress
 - Take online practice exams
 - Access test history and scores
 - Schedule or modify appointments
 - View instructor comments
- The secretary must be able to:
 - Register new students
 - Schedule lessons on behalf of students
 - Modify or cancel appointments
- Instructors must be able to:
 - View upcoming lessons
 - Enter feedback and comments
- Management must be able to:
 - View activity reports
 - Audit reservations and modifications
- IT staff must be able to:
 - Manage user accounts
 - Reset passwords
 - Block or restore access
- The interface must support both desktop and mobile browsers.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- All users will have reliable internet access when interacting with the online features.
- Students will provide accurate personal and payment information.
- Instructors will consistently enter comments after lessons.
- The DMV will provide updates in a format compatible with the system.
- DriverPass will maintain a cloud hosting agreement to support uptime and backups.
- Customers will use valid credit card information during registration.
- The secretary will continue to handle phone-based appointments for users who do not want to schedule online.

Limitations

*Any system you build will naturally have limitations. What limitations do you see in your system design?
What limitations do you have as far as resources, time, budget, or technology?*

- The system will rely fully on internet connectivity; users cannot update data offline.
- Customization of new lesson packages will not be supported in the initial release.
- The project schedule is limited by staff availability and dependencies noted in the Gantt chart.
- Budget limitations may restrict certain advanced features or automation.
- DriverPass depends on external DMV updates, which may not always arrive promptly.
- The system will not include mobile app development; only browser access will be supported at this time.

Gantt Chart

DriverPass Gantt Chart

Project lead: Ebony Anderson

TASK	ASSIGNED TO	PROGRESS	START	END
Initiation				
Collect Requirements	Sam & Jennifer		22-Jan	4-Feb
Create Use Case Diagrams	Sam		11-Feb	18-Feb
Build Activity Diagrams for Each Use Case	Sam		15-Feb	9-Mar
Research User Interface Designs	Toni & Clark		27-Feb	7-Mar
Build Class Diagram	John		1-Mar	9-Mar
Development				
Get Customer Approval	Sam & Jennifer		10-Mar	11-Mar
Build Interface	Development Team		12-Mar	24-Mar
Link DB to Interface	Development Team		24-Mar	3-Apr
Build Business Logic	Development Team / IT		5-Apr	27-Apr
Test System	QA / Entire Project Team		27-Apr	7-May
Delivery & Closure				
Deliver System	Sam & Jennifer		8-May	9-May
Sign-off Meeting	Sam, Jennifer, and Client (Liam & Ian)		9-May	10-May

